

Tildrakizumab for the prophylaxis of graft-versus-host disease after allogeneic hematopoietic stem cell transplantation

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Key Points

- Tildrakizumab can be safely given with a pharmacokinetic profile similar to that in healthy individuals.
- Tildrakizumab resulted in a low incidence of acute GVHD and attenuation of microbiome dominance with potentially pathogenic organisms.

We conducted a phase 1/2 study in which patients undergoing allogeneic hematopoietic stem cell transplantation received tildrakizumab in addition to standard immune suppression with tacrolimus and methotrexate for graft-versus-host disease (GVHD) prophylaxis. A total of 50 patients were enrolled between March 2020 and June 2023 with a median age of 56 years (range, 19-64). All patients received myeloablative busulfan-based conditioning and were transplanted with HLA-matched related or unrelated peripheral blood stem cell grafts. Patients were treated with tildrakizumab on an extended subcutaneous administration schedule for 5 doses, which was well tolerated. The cumulative incidences of grades 2 to 4 and 3 to 4 acute GVHD were 14% (95% confidence interval [CI], 7-28) and 4% (95% CI, 1-16) at day 100, respectively. The incidence of chronic GVHD requiring systemic immune suppression was 52.7% (95% CI, 40.4-68.9) at 12 months. The 1-year probabilities of overall, disease-free, and GVHD-free relapse-free survival were 80% (95% CI, 70-92), 78% (95% CI, 67-90), and 19.3% (90% CI, 11.8-31.4), respectively. Pharmacokinetic analysis revealed the half-life of tildrakizumab at ~28 days without formation of detectable anti-tildrakizumab–neutralizing antibodies. Comparative examination of fecal microbial composition in tildrakizumab and a similarly transplanted cohort treated with tocilizumab prophylaxis demonstrated that both cytokine blockade strategies had a low frequency of enterococcal dominance. We conclude that tildrakizumab resulted in a low incidence of acute GVHD and attenuation of microbiome dominance with potentially pathogenic organisms but did not mitigate the emergence of chronic GVHD as administered on this dosing schedule. This trial was registered at www.clinicaltrials.gov as #NCT04112810.

Introduction

Graft-versus-host disease (GVHD) is the primary complication occurring after allogeneic hematopoietic stem cell transplantation (HSCT) and a major cause of nonrelapse mortality. The pathophysiology

Submitted 12 November 2025; accepted 2 January 2026; prepublished online on *Blood Advances* First Edition 3 February 2026. <https://doi.org/10.1182/bloodadvances.2025019065>.

Deidentified individual participant data that underlie the reported results will be made available 3 months after the publication date. Data are available from the corresponding author, William R. Drobyski (wdrobyski@mcw.edu), on request.

The full-text version of this article contains a data supplement.

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of this disease is characterized by the production of proinflammatory cytokines that perpetuate GVHD through direct cytotoxic effects on host tissues, activation and/or priming of immune effector cells, and differentiation of naive T-cell precursors into proinflammatory T-cell populations (ie, T-helper 1 [Th1] and Th17 cells).¹⁻⁵ The pretransplant conditioning regimen is the initial stimulus for cytokine dysregulation and induces the production of alarmins such as interleukin-33 (IL-33) and IL-1 α ^{6,7} and damage-associated molecular patterns^{8,9} from injured tissue sites. This leads to the activation of antigen-presenting cells resulting in the release of IL-6, IL-12, and IL-23, which drives the development of alloactivated donor T cells along Th1 and Th17 differentiation pathways,^{10,11} while inhibiting the emergence of regulatory T cells,¹² resulting in further inflammatory cytokine production and direct host tissue damage.

IL-23 is a member of the IL-12 family which signals through STAT3 and is secreted by dendritic cells and other APCs, such as macrophages and monocytes.¹³ This cytokine shares a p40 subunit with IL-12 but also contains an IL-23-specific p19 component. The p19/p40 complex binds to the IL-12R β 1 subunit that is also a component of the IL-12 receptor (IL-12R) along with a unique IL-23R subunit that together is present on memory/activated T cells, dendritic cells and macrophages.¹⁴ Preclinical studies using both antibody-based and genetic approaches have revealed that inhibition of the IL-23/IL-23R signaling pathway prevents acute GVHD lethality, attenuates inflammatory cytokine production, and mitigates pathological damage preferentially within the gastrointestinal tract.^{11,15} In addition, although IL-23 induces production of proinflammatory cytokines (ie, IL-6, tumor necrosis factor- α [TNF- α])¹⁶ that are implicated in acute GVHD, it also enhances production of IL-17 that plays a role in the biology of chronic GVHD,¹⁷ positioning IL-23 as a cytokine that has relevance to both acute and chronic GVHD pathophysiology.

Tildrakizumab (Ilumya) is a high-affinity, humanized antibody that binds specifically to the p19 subunit of IL-23¹⁸ and therefore does not interfere with IL-12 signaling. Food and Drug Administration approval was granted based on the efficacy of tildrakizumab in the treatment of moderate-severe plaque psoriasis where responses were superior to those observed with either placebo or etanercept.^{19,20} Notably, the antibody was well tolerated after repetitive dosing, and there was no increased incidence of severe infections or drug-related hypersensitivity when compared with placebo-treated patients,²⁰ indicating that tildrakizumab had an excellent safety profile in this patient population. Given the role of IL-23 for prevention of GVHD in preclinical murine models, the goal of this study was to determine whether blockade of this pathway with tildrakizumab was safe and efficacious for GVHD prophylaxis in humans.

Materials and methods

Study design

This was an investigator-initiated phase 1/2 trial designed to evaluate the safety and efficacy of tildrakizumab administered as adjunctive therapy in addition to tacrolimus and methotrexate for the prevention of GVHD in patients receiving myeloablative allogeneic HSCT. Patients were consented before the start of conditioning. The primary end point of this trial was to determine the

probability of GVHD-free relapse-free survival (GRFS). An event for this outcome was defined as either grade 3 to 4 acute GVHD, systemic therapy required for chronic GVHD, relapse, or death at 12 months. The protocol was approved by the Internal Review Board at the Medical College of Wisconsin with written informed consent obtained from all patients. An independent data and safety monitoring committee oversaw the conduct of the study. This trial was registered at ClinicalTrials.gov (identifier: NCT04112810).

Patient population

Patients were eligible for this trial if they met the following criteria: age ≥ 18 years; a diagnosis of acute lymphoblastic leukemia (ALL) or acute myelogenous leukemia (AML), chronic myelogenous leukemia (CML), myeloproliferative disease, or myelodysplasia (MDS) with $<5\%$ of blasts in the bone marrow; availability of an 8/8 matched sibling or unrelated donor; ejection fraction at rest $>45\%$; estimated creatinine clearance >40 mL/min using the Cockcroft-Gault formula and actual body weight; adjusted diffusing capacity for carbon monoxide $\geq 40\%$ and forced expiratory volume in 1 second $\geq 50\%$; and total bilirubin <3 times and alanine aminotransferase/aspartate aminotransferase <5 times the upper normal limit. Prior autologous stem cell transplant and planned posttransplant maintenance therapy were allowed. Patients with AML and ALL had to be in complete remission at the time of allogeneic HSCT. Patients were excluded if they had had a prior allogeneic HSCT, active central nervous system involvement with malignancy, Karnofsky performance score $<60\%$, uncontrolled bacterial, viral, or fungal infections currently on treatment and with progression of infectious disease or no clinical improvement at time of enrollment, or uncontrolled cardiovascular conditions (ie, congestive heart failure, uncontrolled angina, or ventricular arrhythmias).

Conditioning regimen, GVHD prophylaxis, and anti-infection prophylaxis

Patients receiving myeloablative conditioning were treated with either busulfan ≥ 3.2 mg/kg per day (days -7 to -4) to achieve a steady-state plasma concentration of 900 ng/mL and cyclophosphamide 60 mg/kg per day (days -3 and -2) (BuCY) or busulfan ≥ 3.2 mg/kg per day (days -5 to -2) and fludarabine 50 mg/m² per day (days -5 to -2) (Flu/BU4). All patients received T-cell-replete granulocyte colony factor-stimulated peripheral blood stem cell grafts. For GVHD prevention, tacrolimus was administered IV at a dose of 0.03 mg/kg per day starting day -1 to maintain a level of 5 to 10 ng/mL. Tacrolimus taper was initiated at day $+90$ if no active GVHD with intent to discontinue by day $+180$. Methotrexate was given at the doses of 15 mg/m² IV on day $+1$ and 10 mg/m² IV on days $+3$, $+6$, and $+11$ after hematopoietic stem cell infusion. Antibacterial prophylaxis was with levofloxacin with onset of neutropenia, followed by first-line therapy with cefepime for first fever, unless otherwise dictated by culture results and sensitivities, or drug allergies. Empiric antibiotics were continued until engraftment occurred.

Tildrakizumab administration schedule

Tildrakizumab was administered at a dose of 100 mg subcutaneously on days -1 , $+28 \pm 3$, $+112 \pm 7$, 196 ± 14 , and 280 ± 14 on an inpatient or outpatient basis depending on patient status.

Table 1. Baseline patient and transplant characteristics

Characteristic	N = 50
Age, median (range), y	56 (19-64)
Sex, M/F	27/23
Disease, n (%)	
AML	25 (50)
CR1	22
CR2	3
ALL	4 (8)
B cell, Ph-, CR1	2
B cell, Ph+, CR1	1
T cell, CR1	1
Biphenotypic acute leukemia	1 (2)
Secondary AML	3 (6)
Therapy-related AML, CR1	4 (8)
MDS	4 (8)
Therapy-related MDS	3 (6)
CMML	2 (4)
Myelofibrosis	1 (2)
MDS/myelofibrosis	1 (2)
CML	
Blast crisis	1 (2)
TKI resistant	1 (2)
Donor type, n (%)	
MRD	14 (28)
MUD	36 (72)
Conditioning regimen, n (%)	
Flu/BU4	48 (96)
BU/CY	2 (4)
Graft source, n (%)	
Peripheral blood	50 (100)
ECOG performance status, n (%)	
0	3 (6)
1	30 (60)
2	17 (34)
Karnofsky score, n (%)	
100	1 (2)
90	3 (6)
80	19 (38)
70	20 (40)
60	7 (14)
HCT-CI, n (%)	
0	8 (16)
1	7 (14)
2	9 (18)
3	14 (28)
4	6 (12)
5	2 (4)
6	3 (6)
8	1 (2)

Table 1 (continued)

Characteristic	N = 50
Disease risk index, n (%)	
Low	1 (2)
Intermediate	17 (36)
High	25 (53)
Very high	4 (9)
Missing	3
CMV serostatus, n (%)	
Donor-/recipient-	8 (16)
Donor+/recipient-	2 (4)
Donor+/recipient+	15 (30)
Donor-/recipient+	24 (48)
Donor+/recipient equivocal	1 (2)

CMML, chronic myelomonocytic leukemia; CMV, cytomegalovirus; CR1, Complete Remission 1; ECOG, Eastern Oncology Cooperative Group; HCT-CI, Hematopoietic Cell Transplantation-specific Comorbidity Index; MRD, matched related donor; MUD, matched unrelated donor; TKI, tyrosine kinase inhibitor.

Other detailed methods

All other methods are described in the supplemental Files.

Results

Patient characteristics

Between 24 March 2020, and 7 June 2023, 51 patients were enrolled in the study. One patient withdrew immediately before receiving the first dose of tildrakizumab and was excluded from the analysis. The demographic data for the 50 trial-enrolled and treated patients are detailed in Table 1. The median age of the cohort was 56 years (range, 19-64). 48 patients received Flu/BU4 and 2 received BU/CY. All patients received peripheral blood stem cell grafts from related (n = 14) or matched unrelated (n = 36) donors. Diseases consisted of de novo AML (n = 25), secondary or therapy-related AML (n = 7), ALL (n = 4), MDS (n = 4), therapy-related MDS (n = 3), chronic myelomonocytic leukemia (n = 2), CML (n = 2), myelofibrosis (n = 1), bi-phenotypic acute leukemia (n = 1), and MDS/myelofibrosis (n = 1). The Hematopoietic Cell Transplantation-specific Comorbidity Index (HCT-CI) scores for this patient population were 0 (n = 8), 1 to 2 (n = 16), 3 (n = 14), 4 to 5 (n = 8), and >5 (n = 4). The median Karnofsky score was 70 (range, 60-100). Most patients had high- or very high-risk disease as assessed by the disease risk index.²¹ Donor/recipient cytomegalovirus serostatus is detailed in Table 1. The median follow-up for surviving patients was 934 days (range, 389-1551) as of 30 June 2024.

Engraftment and chimerism

All patients engrafted with a median time to neutrophil recovery (ANC > 500/ul) of 15 days (range, 11-24). One patient never achieved a sustained platelet count >20 000/ul and died 46 days posttransplantation from respiratory failure. In the remaining patients, the median time to platelet engraftment was 17 days (range, 7-33). Chimerism studies conducted on day 28

posttransplantation were available for all 50 patients. Median donor CD3 chimerism was 91% (range, 66%-99%), whereas median CD33 chimerism was 100% (range, 74%-100%).

Tildrakizumab administration

All patients received the first dose of tildrakizumab on day -1. Of 50 patients, 45 (90%) received a second dose at a median of 28 days (range, 25-33), 33 (66%) received a third dose at a median of 112 days (range, 105-123), 20 (40%) a fourth dose at a median of 196 days (range, 182-213), and 12 (24%) a fifth dose at a median of 282 days (range, 269-293). In total, 160 (64%) of the maximum number of 250 doses were administered to the patients in the trial. Furthermore, 66 of the 250 doses (26%) that were not given were due to the occurrence of a predefined GRFS event at some point during the first year that triggered protocol discontinuation. Ten additional doses were omitted due to patient withdrawal from the study (n = 9) or patient refusal (n = 1). In addition, 12 doses were not administered at the discretion of the attending physician due to existing comorbidities (ie, veno-occlusive disease [n = 1], COVID [n = 1], grade 2 acute GVHD on salvage therapy [n = 3], idiopathic pneumonia syndrome [n = 4], ABO incompatibility-induced cytopenias [n = 1], multi-organ failure [n = 1], and neutropenia [n = 1]). Two patients each missed 1 dose due to being treated with steroids for grade 2 acute GVHD but were resumed on tildrakizumab due to discontinuation of steroids before the next dose per protocol guidelines.

Adverse events and infectious complications

Because tildrakizumab was administered multiple times to patients, we identified as an adverse event any incident that occurred within 30 days after a preceding dose of the antibody. The number of grades 3, 4, or 5 nonhematologic adverse events that fulfilled these criteria is found in Table 2. Three patients withdrew from the study after experiencing rash, dizziness, and emesis, respectively. These events occurred after the second dose and were deemed to be not likely due to tildrakizumab. Most adverse events were grade 3 and attributable primarily to side effects often associated with myeloablative allogeneic HSCT. There were 8 grade 4 and 1 grade 5 events, the latter in a patient who died from respiratory failure. Three patients developed veno-occlusive disease of the liver and were treated with defibrotide resulting in resolution of liver abnormalities. Viral infections consisted of cytomegalovirus viremia (n = 4), human herpesvirus 6 viremia (n = 1), COVID (n = 4), BK viruria (n = 1), and rhinovirus (n = 1). Three patients had bacterial blood stream infections (*Leptotrichia*, *Fusobacterium*, and *Streptococcus mitis*), and 1 patient had *Clostridium difficile* toxin detected from the stool all within the first 100 days.

Acute and chronic GVHD

The cumulative incidence of grades 2 to 4 acute GVHD at days 100 and 180 was 14.0% (95% confidence interval [CI], 7.0-27.8) and 18% (95% CI, 10.0-32.5), respectively (Figure 1A). The incidence of grades 3 to 4 acute GVHD at these same time points was 4% (95% CI, 1.0-15.6) and 4% (95% CI, 1.0-15.6), respectively (Figure 1A). Seven patients developed grade 2 acute GVHD, which involved the skin in 4 patients, the lower gastrointestinal (GI) tract in 2 individuals, and the skin and GI tract in the last patient. There were 2 cases of grade 3 acute GVHD involving

Table 2. Tildrakizumab serious adverse events

	Grade 3	Grade 4	Grade 5
Gastrointestinal			
Oropharyngeal mucositis	36	3	
Nausea	12		
Anorexia	8		
Diarrhea	9		
Elevated transaminases	7	1	
Colitis/typhlitis	2		
Hepatic veno-occlusive disease	1	2	
Abdominal pain	3		
Duodenal ulcer	1		
Cardiovascular			
Arrhythmia	2		
Hypertension	13		
Hypotension	2		
Chest pain	2		
Respiratory			
Hypoxia	5		
Dyspnea	5		
Idiopathic pneumonia syndrome	1	1	
Respiratory failure			1
Metabolic			
Hyperglycemia	6		
Hyponatremia	3		
Hypokalemia	2		
Hypocalcemia	3		
Hypophosphatemia	1		
Hypermagnesemia	1		
Infectious			
Febrile neutropenia	11		
Sepsis	4		
Perineal abscess	1		
Renal			
Acute kidney injury	3		
Hematuria	2		
Neurologic			
Confusion/delirium	1	1	
Syncope	3		
Hematologic			
Hemolysis	1		

Adverse events consisted of all episodes that occurred within 30 days after any dose of tildrakizumab administration. Nonhematologic adverse events were graded according to the revised National Cancer Institute Common Terminology Criteria for Adverse Events, version 5.

the lower GI tract in 1 patient and the skin, liver, and GI tract in the second recipient within the first 6 months. The cumulative incidence of chronic GVHD which required systemic immune suppression was 18% (95% CI, 10.0-32.5) and 52.7% (95% CI, 40.4-68.9) at 6 and 12 months, respectively (Figure 1B). Of the 26 patients who required systemic immunosuppressive therapy within

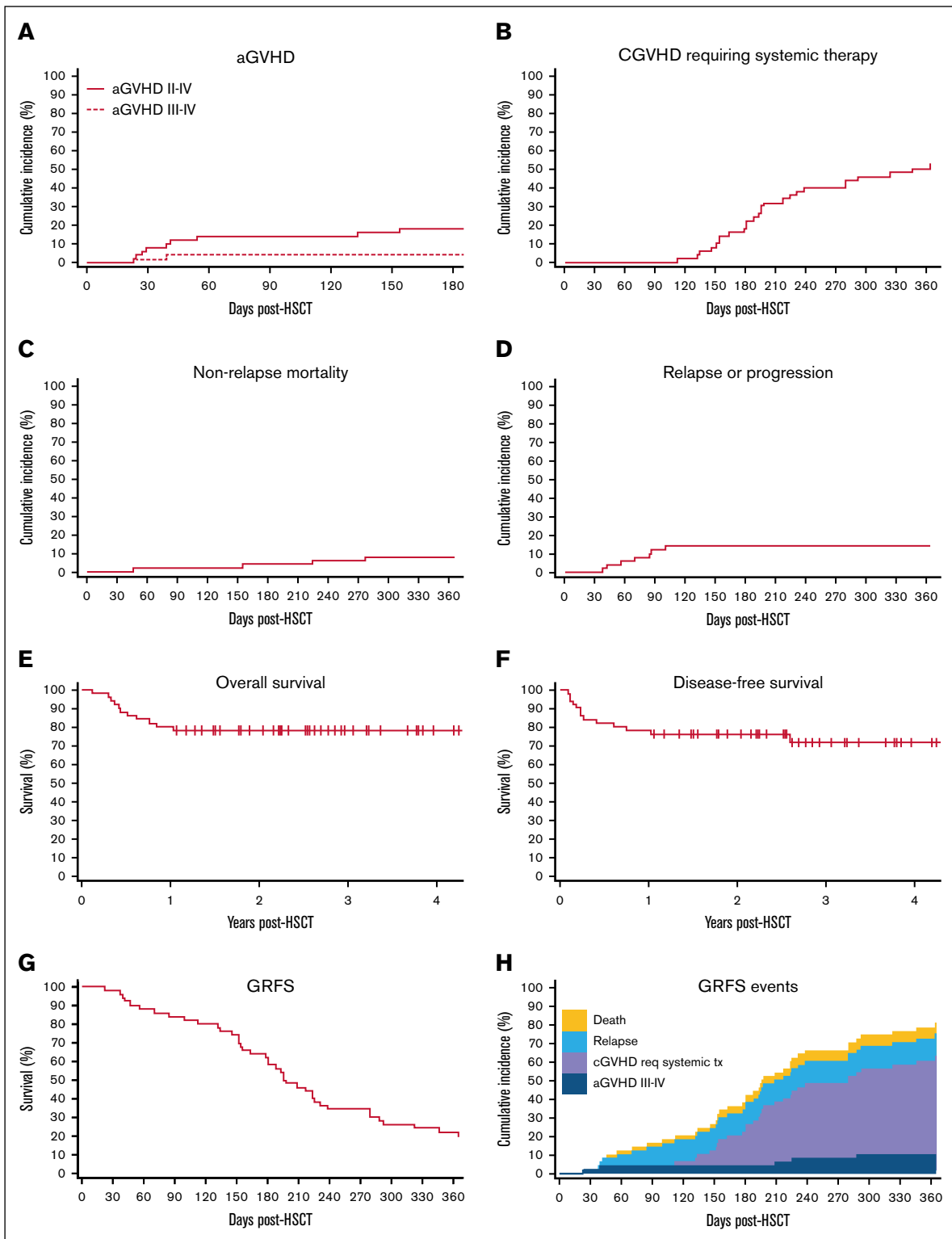


Figure 1. GVHD, transplant-related mortality, relapse, GRFS, and survival. (A) Cumulative incidence of grades 2 to 4 and grades 3 to 4 acute GvHD. (B) Cumulative incidence of chronic GvHD requiring systemic immunosuppressive therapy, (C) nonrelapse mortality, or (D) relapse. (E) Probability of overall survival, (F) disease-free survival, or (G) GRFS. (H) Cumulative incidence of individual GRFS events.

the first year, there were 5 with mild GVHD, 15 with moderate, and 6 with severe chronic GVHD as determined using the National Institutes of Health consensus criteria.²¹

Transplant-related mortality, relapse, GRFS, and survival outcomes

Transplant-related mortality was 8.0% (95% CI, 3.1-20.5) at 12 months (Figure 1C) and was attributable to idiopathic pneumonia syndrome (n = 2), respiratory failure (n = 1), and pneumonia/acute respiratory distress syndrome (n = 1). The 1-year cumulative incidence of relapse was 14% (95% CI, 7.0-27.8) and occurred 38 to 118 days posttransplantation (Figure 1D). Relapse occurred in a total of 7 patients; 3 with p53-mutated AML (n = 2) or MDS (n = 1) and 1 each with monosomal karyotype AML, CML in blast crisis, secondary AML, and minimal residual disease-positive ALL. Overall and disease-free survival were 80% (95% CI, 69.7-91.9) and 78% (95% CI, 67.3-90.4), respectively, at 1 year (Figure 1E-F).

The probability of GRFS at 1 year was 19.3% (90% CI, 11.8-31.4) (Figure 1G). The null hypothesis that the probability of GRFS at 12 months is 20% was therefore not rejected ($P = .55$). A competing-risk survival analysis demonstrated that the primary GRFS triggering event was chronic GVHD requiring systemic immunosuppression (52.7%, [95% CI, 40.4-68.9]), whereas grades 3 to 4 acute GVHD (10.0%, [95% CI, 4.4-23]), relapse (12%, [95% CI, 5.7-25.4]), and death (6.0%, [95% CI, 2.0-18.0]) were less common events that triggered discontinuation of tildrakizumab (Figure 1H).

PK analysis

Peripheral blood samples were collected from patients before tildrakizumab administration (baseline) and at 30 ± 2 , 60 ± 7 , and 90 ± 14 days posttransplantation. In all cases except one, the day +30 pharmacokinetic (PK) sample was collected before the second tildrakizumab treatment was administered to patients; therefore, this measurement constituted a trough value after the first dose. The concentration of tildrakizumab in individual patients over time is depicted in Figure 2A and summarized in box plots at the nominal times of 30-, 60- and 90-days posttransplantation (Figure 2B). The calculated mean concentration of tildrakizumab at day 30 was 3860 ng/mL, which rose to 7370 ng/mL (day 60) before declining to 3516 ng/mL (day 90) ~2 months after the prior dose (Table 3). A population PK model was used to derive the predicted concentration data and calculate relevant PK parameters (see "Materials and methods"). A noncompartmental model was then used to determine the time to maximal concentration (T_{max}), maximal concentration (C_{max}), half-life ($t_{1/2}$), and area under the curve of tildrakizumab (Table 4). This analysis revealed that the T_{max} of tildrakizumab was 6 to 7 days after subcutaneous administration. In addition, the geometric mean of the half-life, which was calculated after the first and second doses of antibody treatment, was similar and ~28 days. To evaluate whether patients were able to mount an immune response against tildrakizumab, we also performed screening for the detection of anti-tildrakizumab antibodies in 169 serum samples. There were no cases of anti-tildrakizumab antibody formation as assessed by an immunoassay that involved prespecified screening and confirmatory steps.

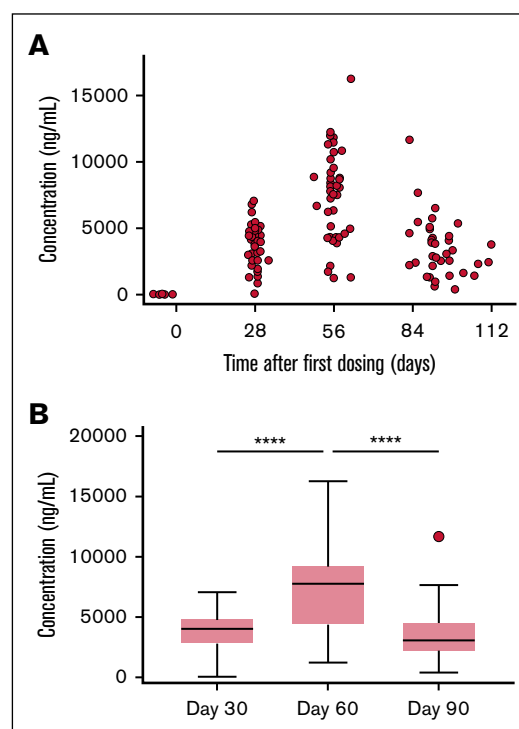


Figure 2. PK analysis of tildrakizumab. (A) Tildrakizumab concentration vs time. Individual points represent the serum values in each patient at a given time point posttransplantation. (B) Box plot of tildrakizumab concentration over time. The solid line within the box indicates the median concentration of tildrakizumab. The bottom and top edges of the box represent the 25th and 75th percentiles. The lower and upper whiskers are the smallest and largest points within 1.5 interquartile points below or above the 25th and 75th percentiles, respectively. The point outside of the box at day 90 represents an outlier. **** $P < .0001$.

Serum cytokine analysis

Proinflammatory cytokines which have been implicated in the pathophysiology of GVHD (ie, IL-23, IL-17, IL-6, IFN- γ , granulocyte-macrophage colony-stimulating factor, and TNF- α)¹⁻⁵ were evaluated at multiple time points that spanned the period of tildrakizumab administration. The measured values of each cytokine in individual patients are found in Figure 3A and demonstrate that IL-23 concentrations were substantially higher than other cytokines at all time points. To determine whether there were statistically significant changes within in-patient values over time, a mixed linear effects model was applied to log-transformed data for each cytokine. Results of this analysis revealed that IL-23 levels were relatively stable in the first 6 months (Figure 3B; supplemental Table 1). An early increase in IL-6 levels was observed followed by return to baseline values as has been noted in prior studies.^{26,28} There were also increases in other cytokines, generally within the first 60 days, but the incremental increases were modest (supplemental Table 1).

Comparative microbiome analysis between tildrakizumab- and tocilizumab-treated patients

Antibody blockade of the IL-6 and IL-23 signaling pathways in preclinical models has both been found to protect recipients from GVHD, particularly in the GI tract.^{11,12} In addition, we previously

Table 3. Tildrakizumab Concentrations

Interval	N	Mean ng/mL	SD ng/mL	CV%	Min ng/mL	Median ng/mL	Max ng/mL	Range ng/mL
Day 30	45	3859.96	1565.84	40.6	70.53	4057.71	7 073.75	7 003.22
Day 60	42	7369.58	3327.79	45.2	1271.89	7788.76	16 260.43	14 988.53
Day 90	35	3515.89	2237.43	63.6	391.24	3053.51	11 704.21	11 312.97

Descriptive statistical results of the observed tildrakizumab concentration data collected at the nominal times of baseline, day 30, day 60, and day 90.
CV, Coefficient of variation; Max, Maximum; Min, Minimum; SD, Standard Deviation.

reported that administration of the anti-IL-6R antibody, tocilizumab, for GVHD prophylaxis in patients was associated with reduced lower GI tract GVHD, preservation of microbial diversity, and an attenuation of enterococcal domination of the intestinal microbial community.²² To assess the effect of tildrakizumab administration on the microbiome, we profiled fecal microbial composition in patients who were treated with this antibody and compared results against a prior published cohort of patients who received tocilizumab for GVHD prophylaxis.²² Patients in both groups received the same busulfan-based conditioning regimen, standardized GVHD prophylaxis with a backbone of tacrolimus and methotrexate, stem cell graft source, antibiotic prophylaxis, and antibiotic therapy for first fever. The microbiome taxonomic profiles were generated by 16S rRNA gene sequencing of 82 stool samples collected from 19 patients who received tocilizumab (TOCI) and 214 samples from 47 patients that were treated with tildrakizumab (TILDRA).

As expected, the baseline fecal compositions were relatively diverse communities enriched for common strictly anaerobic commensal organisms. Baseline beta diversity differed significantly between the TILDRA and TOCI cohorts (permutational multivariate analysis of variance, $P = .001$), and PCoA ordination revealed separation by cohort along axis 1 (Figure 4A). Stool alpha diversity was significantly higher in the TILDRA cohort than the TOCI cohort, both at baseline and to the 4 weeks after transplantation (Figure 4B; supplemental Table 3). Examination of temporal alterations in specific taxonomic groups of interest (ie, *Akkermansia*,^{23,24} *Blautia*,²⁵ *Escherichia*,²⁶ and *Flavonifractor*²⁷) that have relevance in the pathophysiology of GVHD revealed no difference in the fold-change trajectories over time for any of these groups (Figure 4C). In a univariate model describing the community differences at baseline, TILDRA patients were enriched for the

abundance of several butyrate-producing anaerobic genera, including *Blautia*, *Flavonifractor*, *Mediterraneibacter*, and *Flintibacter*, whereas the TOCI patients were enriched for the genera *Escherichia* and *Faecalitalea* (Figure 4D; supplemental Table 4). In the posttransplant longitudinal model, the *Escherichia* and *Flavonifractor* differences found at baseline were found again to a greater degree (supplemental Table 5). The incidence of dominated samples was lower in the TILDRA cohort and demonstrated less *Akkermansia* domination than was observed among the TOCI samples (Figure 4E). Notably, neither cohort had evidence of domination with *Enterococcus*, which is a pathogen associated with increased GVHD-related mortality.²⁸ Differences in individual fecal microbial composition are illustrated in stacked bar plots of taxonomic composition where firmicutes comprised most of the bacterial population in the TILDRA cohort, whereas *Akkermansia* (light blue) and sporadic *Escherichia* (dark red) dominations were more often observed in the TOCI patient population (Figure 4F-G).

Discussion

In this study, we evaluated the safety and efficacy of tildrakizumab in addition to standard immune suppression with tacrolimus and methotrexate for the prevention of acute and chronic GVHD occurring after myeloablative allogeneic HSCT. The foundation for this trial derived from preclinical studies that demonstrated blockade of IL-23 signaling using genetic- or antibody-based approaches significantly reduced GVHD lethality in mice and preferentially protected the GI tract from immune-mediated attack.^{11,15} Results of this study demonstrated that tildrakizumab was generally well tolerated and could be repetitively administered without any discernible cumulative toxicity. Furthermore, there was no evidence that tildrakizumab administration led to an increase in infectious complications. With respect to GVHD prevention, there was a low incidence of grades 2 to 4 (14%) and 3 to 4 (4%) acute GVHD at day 100 when compared with other reports in which tacrolimus and methotrexate were used as sole GVHD prophylaxis,²⁹⁻³¹ suggesting that tildrakizumab has efficacy in attenuating acute GVHD severity.

Prior studies using cytokine-blockade strategies to mitigate GVHD administered agents over relatively short periods of time in the more immediate posttransplant setting.³¹⁻³⁴ In this study, we gave a more protracted course of treatment to determine whether this therapeutic approach could also affect the emergence and severity of chronic GVHD. To that end, we extrapolated the tildrakizumab dose and administration schedule from prior clinical studies in which this antibody had been administered repetitively in the first year to patients with psoriasis.¹⁹ In this population, clinical efficacy was similar at 100 and 200 mg doses, but inferior at 25 mg, which

Table 4. Pharmacokinetic Parameters

Parameter	First dose	Second dose
T _{max} (d)	7.00 (4.5, 10.0)	5.98 (3.86, 8.11)
C _{max} (ng/mL)	7 260 (9.1%)	11 000 (13.2%)
t _{1/2} (d)	27.9 (32.7%)	28.1 (31.9%)
AUC _{D0-D28} (ng d/mL)	164 000 (13.0%)	–
AUC _{D28-D112} (ng d/mL)	–	448 000 (31.5%)
C _{avg} (ng/mL)	–	5 330 (31.5%)

T_{max} is presented as the median (minimum, maximum). C_{max}, t_{1/2}, AUC, and C_{avg} are presented as the geometric mean plus the coefficient of variation in parentheses.
AUC, area under the curve, C_{avg}, Average concentration.

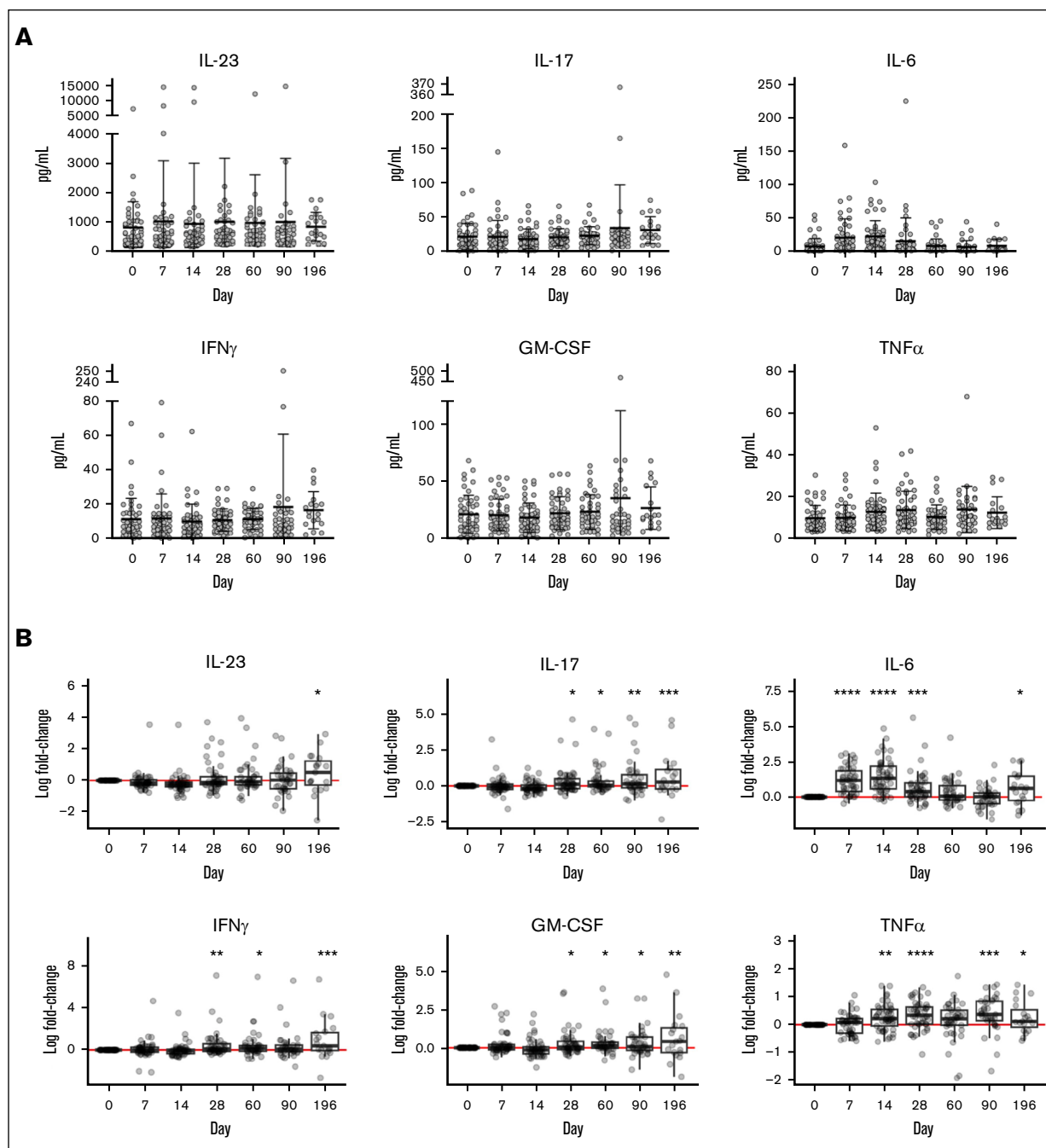


Figure 3. Cytokine levels in patients who received tildrakizumab for GVHD prophylaxis. (A) Scatterplot denoting concentration of specified serum cytokines. Samples were obtained on days 0 (n = 49), +7 (n = 49), +14 (n = 50), +28 (n = 48), +60 (n = 41), +90 (n = 37), and +196 (n = 19). Patients who developed one of the GRFS end points were taken off trial, and samples from these patients were not assessed beyond the time point at which the GRFS event was triggered. Data are illustrated as the mean $\pm$ SD. (B) Box plots depicting log fold change from baseline of specified cytokines over time. The red line illustrates the NULL hypothesis of no change from baseline for reference. The solid line within the box indicates median log fold change. The lower and upper edges of the box denote 25th and 75th quantiles, whereas the lower and upper whiskers are drawn out to the most extreme observations within 1.5 times the interquartile range from the upper and lower edges. * $P < .05$; ** $P < .01$; *** $P < .001$; **** $P < .0001$.

thereby guided our selection of 100 mg as the dosage of choice. However, because the metabolism of tildrakizumab in patients with allogeneic HSCT is unknown, we also performed PK studies to define the pharmacological parameters of the antibody in this patient population. We observed that the PK levels of

subcutaneously administered tildrakizumab were similar to that in healthy subjects who received either 50 or 200 mg doses.³⁵ In these individuals, the half-life was noted to be ~25 days, which was comparable to the average of ~28 days that we observed in our patient population. The time to maximal antibody concentration

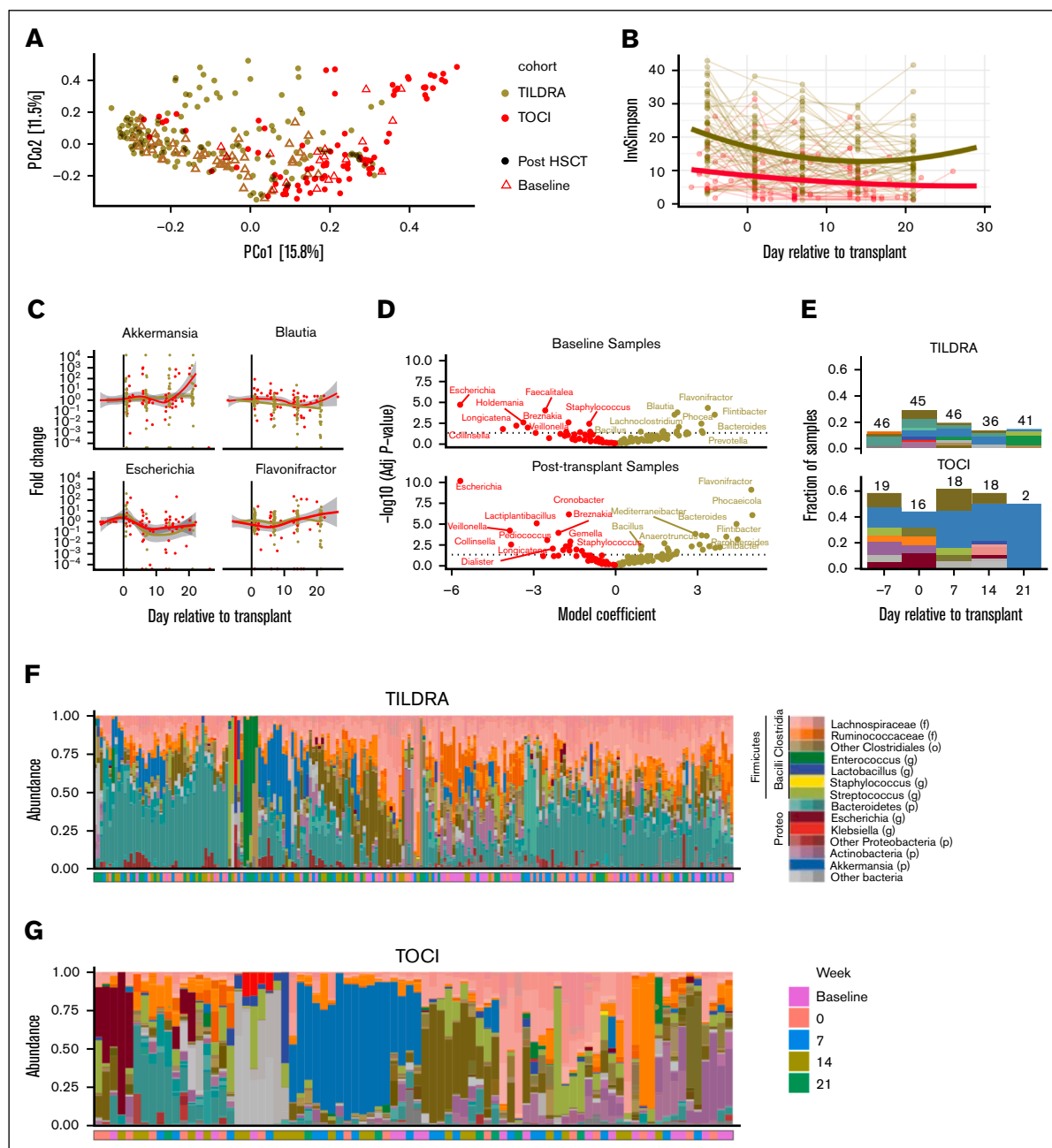


Figure 4. Microbiome analysis of TILDRA and TOCI stool samples. (A) Principal coordinate analysis of all samples. Baseline samples (open triangles) are significantly different between TILDRA and TOCI patients (permutational multivariate analysis of variance, $P = .001$). (B) Alpha diversity by patient and with group estimates according to generalized estimating equation (GEE) model. See supplemental Table 3 for summary of the GEE model. Each point is a fecal sample; the curves are smoothed averages. (C) Fold change in relative abundances, compared with baseline, of selected taxa are plotted over time. Each point represents a sample; curves are LOESS smoothed averages and shaded areas reveal 95% CIs. (D) Differential abundance between TILDRA and TOCI cohorts in baseline-only model (top) and posttransplant longitudinal model (bottom); positive coefficient indicates enrichment in TILDRA cohort. There were 65 samples in the baseline group (46 from the TILDRA group and 19 from the TOCI group), whereas there were 227 samples in the posttransplant group (164 from the TILDRA group and 63 from the TOCI group). (E) Taxa contributing to domination events in samples in 7-day bins from the 2 cohorts. Total stacked bar height reveals the proportion of samples with dominated microbiome (any amplicon sequence variant with relative abundance exceeding 30%), illustrated by week and colored by dominating taxon using the color scheme described in panel F. Gray number indicates the number of patients with evaluable samples in that week. (F-G) Genus-level composition of samples in TILDRA (F) and TOCI (G) cohorts and color coded according to the scheme described in Peled et al.⁴⁹ Samples are ordered by hierarchical clustering; week of collection is illustrated beneath as a heat map. In all panels, the TILDRA cohort includes 210 samples from 46 patients and TOCI cohort consists of 82 samples from 19 patients. Figure 4E,G adapted from Chhabra et al²² with permission.

in the serum (ie, 6-7 days) was also similar to that reported in healthy individuals.³⁵ Notably, anti-tildrakizumab antibody testing yielded negative results in all patients, indicating that tildrakizumab did not induce an immunogenic response that could have compromised efficacy.

Examination of inflammatory cytokines in the serum revealed that IL-23 levels were significantly higher than other cytokines (ie, IL-17, IL-6, IFN- γ , granulocyte-macrophage colony-stimulating factor, and TNF- α) that are known to play a role in GVHD pathophysiology. Because tildrakizumab binds to the p19 subunit of IL-23, it is possible that this may have resulted in higher circulating IL-23 concentrations due to inability of tildrakizumab-bound IL-23 to bind to its native receptor. However, the fact that IL-23 levels remained persistently high at days 90 and 196, which was 2 to 3 months after the prior dose of tildrakizumab had been administered, would indicate that this does not fully explain this observation given that the half-life of the antibody is ~28 days. Rather, we would conclude that IL-23 seems to be systemically dysregulated during GVHD as has been reported in other inflammatory diseases such as systemic lupus erythematosus and ankylosing spondylitis.^{36,37}

Preclinical murine models have demonstrated a pathophysiological role for IL-17 in sclerodermatous skin and bronchiolitis obliterans, which are well recognized manifestations of chronic GVHD in humans.³⁸ Moreover, an increased population of T cells that produce IL-17 has been found to be increased in the peripheral blood and tissue sites (ie, skin and liver) of humans with this disease.³⁹ The differentiation of naive T cells into IL-17-producing CD4⁺ T cells (Th17 cells) is primarily dependent on exposure to TGF- β and IL-6,⁴⁰⁻⁴² although IL-21 along with TGF- β can also induce differentiation of Th17 cells.⁴³ IL-23 is not required for Th17 differentiation but functions to stabilize and sustain the lineage commitment of developing Th17 cells.⁴⁴ Furthermore, blockade of the IL-23 signaling pathway has been found to mitigate disease severity in IL-17-dependent pathological disorders.^{40,45,46} Therefore, we postulated that blockade of IL-23 signaling might attenuate the incidence and severity of chronic GVHD due its inhibitory effects on IL-17 production.

Unfortunately, we did not observe a salutary effect from tildrakizumab administration on the incidence of chronic GVHD requiring systemic immunosuppression. In fact, this was the primary explanation for the low probability of GRFS at 1 year, as the incidence of chronic GVHD requiring systemic therapy was responsible for nearly two-thirds of the total number of GRFS events. The fact that tildrakizumab administration did not lead to a reduction in chronic GVHD could be due to several factors. One is that the dosing interval of 84 days after administration of the second dose may have been too long to maintain an effective IL-23 blockade given that the half-life of tildrakizumab was calculated to be ~28 days. Thus, tildrakizumab levels may have been insufficient during the window within which chronic GVHD develops, raising the question as to whether more frequent administration may have been more efficacious and highlighting a potential challenge when using cytokine blockade strategies that require protracted administration. Second, because GVHD is a multisystemic disease, and arguably more inflammatory than psoriasis, it is conceivable that a higher dose may have been more effective, although the fact that the incidence of acute

GVHD was low would argue against this premise. Finally, IL-17 may not be the sole or primary driver of chronic GVHD pathophysiology, and other nonredundant cytokines and/or inflammatory pathways may drive pathology.⁴⁷

Loss of microbial diversity and the expansion of potentially pathogenic organisms such as *Enterococcus* that are associated with acute GVHD can serve as predictors for subsequent infectious complications and overall mortality.^{28,48,49} Whether blockade of inflammatory cytokines designed to reduce GVHD severity can also attenuate microbial dysbiosis is not known. To address this question, we evaluated the fecal microbial composition in patients from this study, including those from a prior report in which patients were administered prophylactic tocilizumab.²² Notably, in both studies, patients had similarly low incidences of grades 2 to 4 and 3 to 4 acute GVHD, were enrolled with the same trial inclusion criteria, and received identical antibiotic prophylaxis and treatment regimens. Therefore, many of the potentially confounding factors that exist when stool samples from different institutions are comparatively analyzed were not present which allowed us to evaluate the effect of 2 different IL-6 family cytokine blockade strategies on fecal microbial composition. We observed that baseline diversity was higher in tildrakizumab-treated patients and that principal coordinate plots demonstrated 2 largely nonoverlapping populations. The reason for this disparity is not clear, although the different DNA isolation procedures may have been a contributing factor.⁵⁰ The major observation from this analysis was that in both cohorts there was little evidence of dominance of *Enterococcus* which is a hallmark of GVHD-induced dysbiosis.⁴⁴ Only in the TOCI cohort did *Akkermansia*, whose role in intestinal and GVHD biology has yet to be fully defined,^{22,51-53} emerge as a dominant organism. Thus, administration of tocilizumab or tildrakizumab early posttransplantation was associated with an attenuated expansion of organisms associated with GVHD dysbiosis.

In summary, this study demonstrated that tildrakizumab can be safely given to allogeneic HSCT recipients with no evidence of cumulative toxicity on repetitive dosing. Addition of tildrakizumab to standard immune suppression resulted in a low incidence of acute GVHD but had no discernible impact on chronic GVHD when administered on this dosing schedule. Examination of the fecal microbiome revealed little evidence of domination with potentially pathogenic organisms. Thus, GVHD prophylaxis with tildrakizumab represents a possible therapeutic approach for reducing acute GVHD and remodeling the intestinal microbiome. Moreover, tildrakizumab could serve as an adjunctive treatment to myeloablative posttransplantation cyclophosphamide-based regimens⁵⁴ to further reduce the severity of acute GVHD.

Acknowledgments

This research was supported by funding from R01 HL 153192 (L.R., A.S., N.S., B.J., and W.R.D.), K08 HL143189 (J.U.P.), and the Memorial Sloan Kettering Cancer Center Cancer Core Grant National Cancer Institute P30 CA008748 (J.U.P.). N.N.S. is a scholar in clinical research of The Leukemia & Lymphoma Society. This research was also supported by the William Randolph Hearst Foundation, Society of Memorial Sloan Kettering Cancer Center, Paula and Rodger Riney Multiple Myeloma Research Initiative, and The V Foundation (J.U.P.). Support for

the clinical trial was provided by Sun Pharmaceuticals Industries, Princeton, NJ, and consisted of provision of tildrakizumab for all patients enrolled in the trial and payment for costs accrued from the Medical College of Wisconsin (MCW) Cancer Center Clinical Trials Office, Milwaukee, WI. Data management and regulatory support were provided by the MCW Clinical Trials Office. Support for this trial was also provided by the Cell Therapy and Biostatistical Shared Research Cores of the MCW Cancer Center.

Authorship

Contribution: L.R. designed and wrote the clinical study, provided clinical care of patients, analyzed the data, and edited the paper; S.F., G.Y., and L.S. assisted in clinical trial management and acquisition of data; A.S. and L.E.R. provided statistical analysis of clinical and correlative data; K.P., B.J., and T.K. processed and stored serum samples for cytokine analysis; S.A., S.C., M.H., W.L., and N.N.S. enrolled patients on the trial, provided clinical care, and edited the paper; N.S., J.H., A.G., N.W., and J.U.P. performed DNA sequencing and microbiome computational analyses, and edited the paper; and W.R.D. conceived of and designed the study, provided clinical care, analyzed the data, and wrote the paper.

Conflict-of-interest disclosure: S.A. reports receiving research funding from Incyte Inc and AltruBio; and consulting and honoraria from Sanofi and Incyte. M.H. reports receiving research support/funding from ADC Therapeutics and Spectrum; serving as a consultant for Genmab, Clustered Regularly Interspaced Short Palindromic Repeats, AlloVir, Bristol Myers Squibb (BMS), Caribou, Autolus, Forte Biosciences, Kite, and Daiichi Sankyo; and serving on the speaker's bureau for AstraZeneca, ADC Therapeutics, BeiGene, Kite, and Sobi (all unrelated to this study). S.C. reports receiving honoraria from Sanofi, Sobi, Ascentage Pharma, BMS, Pfizer, and Legend Biotech; and institutional research

funding from C4 Therapeutics, CARsgen, Cynata Therapeutics, Johnson & Johnson, AstraZeneca, Incyte, AbbVie, Cullinan Therapeutics, and Takeda. N.N.S. reports participation on advisory boards and/or consultancy for Gilead-Kite, BMS-Juno, Miltenyi Biomedicine, Lilly Oncology, Incyte, AbbVie, Cargo, BeiGene, Kite, Allogene, AstraZeneca, BMS, Ipsen, Genentech, and Galapagos; receiving research funding from Lilly Oncology, Genentech, and Miltenyi Biomedicine; receiving travel support from Lilly Oncology and Miltenyi Biomedicine; and serving on the scientific advisory board for Tundra Therapeutics. J.U.P. reports receiving research funding, intellectual property fees, and travel reimbursement from Seres Therapeutics; receiving consulting fees from Da Volterra, CSL Behring, Crestone Inc, MaaT Pharma, Canaccord Genuity, Inc, and RA Capital; serving on an advisory board of and holding equity in Postbiotics Plus Research and Prodigy Biosciences; and having filed intellectual property applications related to the microbiome (reference numbers 62/843849, 62/977908, and 15/756845). The Memorial Sloan Kettering Cancer Center has financial interests relative to Seres Therapeutics. The remaining authors declare no competing financial interests.

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